THE UNIVERSITY OF TEXAS AT DALLAS

ECS 1100

Sector 5: Advanced Manufacturing, Biomanufacturing & Supply Chain

Week 12 · Industry panels Nov 10–12, 2026

Erik Jonsson School of Engineering and Computer Science

Learning objectives

By the end of the week, you will be able to:

1. Describe how a product goes from design to delivery
2. Explain what makes biomanufacturing different
3. Identify where your major fits in this sector
4. Explain how a breakdown in production or supply affects the whole system

Source: ECS 1100 Fall 2026 course schedule, Week 12 sector outcomes

How everyday things get made



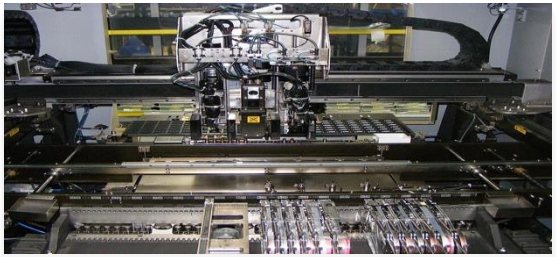
A pickup truck

- Parts from suppliers
- Robots weld and paint
- Chips and software added
- Shipped by rail and truck



A vaccine

- Grow living cells
- Purify the product
- Fill sterile vials
- Test every batch



A phone

- Chips made in a fab
- Cameras inspect boards
- Final test
- Shipped worldwide

Source: Photos via Wikimedia Commons: "KUKA Industrial Robots IR" by Mixabest (CC BY-SA 3.0); "COVIran Barekat vaccine production 01" by Hamed Jafarnejad (CC BY 4.0); "Pick and place internals of surface mount machine" by Peripitus (CC BY-SA 3.0). Icons: Font Awesome

One sector, three branches



Advanced manufacturing

Making things with new technology: robots, sensors, data and new materials.

Examples

Cars, aircraft, chips, medical devices



Biomanufacturing

Using living cells and microbes as the factory. Most modern medicines are made this way.

Examples

Vaccines, antibodies, insulin



Supply chain

Sourcing, buying and moving goods from raw material to the customer.

Examples

Suppliers, warehouses, rail, ports

Source: PCAST/STPI (2010); White House OSTP; CSCMP (accessed Nov 1, 2026). Photos (Wikimedia Commons): Fadal CNC machines by Marcin Wichary (CC BY 2.0); "BTEC Bioreactors" by RickLawless (CC BY 3.0); Port of Los Angeles by Downtowngal (CC BY-SA 4.0)

By the numbers

United States

12.65M manufacturing jobs, Sep 2026

9.7% of U.S. GDP is manufacturing value added (Q2 2026)

3.8M workers needed by 2033; up to 1.9M jobs could go unfilled

Texas

$332B manufacturing output, 11.4% of state GDP

~1M manufacturing jobs, #2 in the nation; average wage about $97,500

#1 exporting state for 24 years in a row

Source: FRED series MANEMP (Sep 2026) and VAPGDPMA (Q2 2026); Manufacturing Institute and Deloitte (Apr 3, 2024); Office of the Texas Governor (Oct 1, 2026)

From design intent to delivery



Design intent	Process planning	Production	Quality	Logistics	Delivery
What it must do, CAD, materials, cost targets	Tooling, line layout, cycle time, validation plan	Machines, robotic cells, PLCs, bioreactors	Inspection, machine vision, SPC, release testing	Planning, inventory, warehouses, freight	Customer, service, data back to design
Majors often here	Majors often here	Majors often here	Majors often here	Majors often here	Majors often here
ME, MSE, BMEN, EE	ME, SYSE, MSE	ME, CE, EE, BMEN	DS, CS, BMEN, MSE	SYSE, DS, SE	SYSE, SE, CTP

Software and data run through every stage. CS and SE build the manufacturing execution systems (MES) and digital thread that track each part; Cybersecurity Technology & Policy protects the factory network (OT) that runs the machines.

Source: ECS 1100 Sector 5 outcomes and degree-program matrix (D. Burleson, Fall 2026)

The biomanufacturing branch



Upstream	Harvest	Downstream	Fill-finish	QC release
Grow cells in a bioreactor; feed, oxygen, pH and temperature held tight	Separate cells from the liquid that holds the product	Purify the product, mostly by chromatography and filtration	Fill sterile vials or syringes in a clean room	Test every batch; quality signs before it ships

Why a living process is harder to control

- Cells vary from batch to batch, unlike a machined part
- One contamination can end a batch worth millions
- You cannot inspect quality in at the end, so every step is validated and recorded
- The record is part of the product: no record, no release

The rules: current Good Manufacturing Practice (cGMP)

21 CFR 210/211	Drugs
21 CFR 600/610	Biologics such as vaccines
21 CFR 820 (QMSR)	Medical devices; aligned with ISO 13485 since Feb 2, 2026

Source: UF Biotility, biomanufacturing process overview (Sep 18, 2025); 21 CFR Parts 210/211, 600/610 and 820 (QMSR, effective Feb 2, 2026), eCFR and FDA (accessed Nov 1, 2026)

Supply chain: the system between the factories

Suppliers → **Inbound freight** → **Factory** → **Warehouse** → **Customer**

Data and forecasts flow back the other way: demand plans tell factories and suppliers what to make next.

255,100

logisticians in the U.S.

$82,320

median logistician pay

18%

projected job growth for logisticians, much faster than average

6.6M

U.S. jobs in transportation and warehousing



North Texas is a logistics hub. DFW Airport handled $62.6B in trade in 2024. AllianceTexas in Fort Worth supports about 136,780 jobs, and its BNSF rail facility handles about 1.1M container lifts a year.

Source: U.S. BLS Occupational Outlook Handbook and CES (accessed Nov 1, 2026); DFW Airport; Hillwood AllianceTexas. Photo: intermodal containers, Knoxville TN, by Lee Ann Ratledge (CC BY 4.0), Wikimedia Commons

Where automation, robots and vision enter

Production

Robotic cells

Welding, painting, assembly and handling; programmed through PLCs

Quality

Machine vision

Cameras and AI check every part instead of a sample

Logistics

Mobile robots

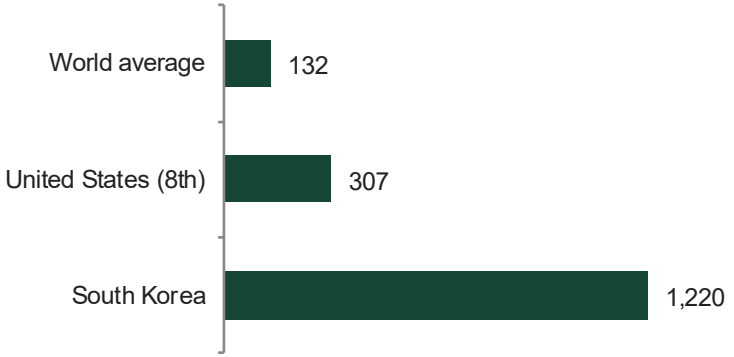
Move material through plants and warehouses

All stages

Digital twin

A live software model of the line used to test changes before making them

Robot density, robots per 10,000 workers



38,000 industrial robots installed in the U.S. in 2025 (+11%), 13,500 of them in auto plants. China installed 295,000 in 2024, against 34,200 in the U.S.

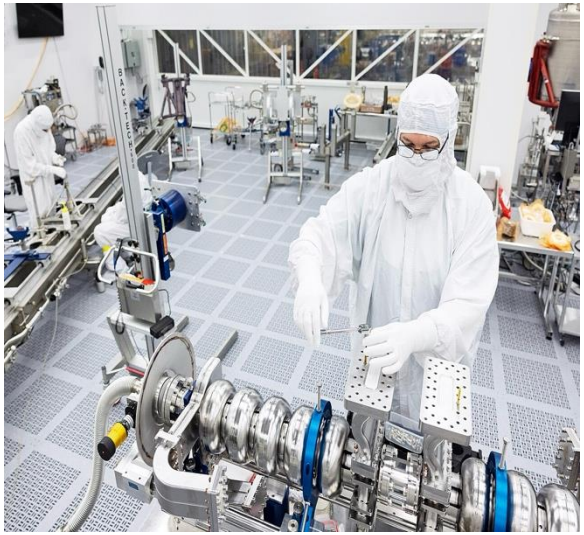
Source: International Federation of Robotics, World Robotics 2025 and 2026 releases; NIST digital twin overview (Feb 13, 2026)

All ten Jonsson School programs have a role

ME	**Mechanical Engineering**	PRIMARY	Process and tooling design, additive manufacturing, bioreactor and fill-line equipment
BMEN	**Biomedical Engineering**	PRIMARY	Biomanufacturing: culture, purification, fill-finish, sterile process, GMP
SYSE	**Systems Engineering**	PRIMARY	Production systems, logistics, quality systems, operations research
DS	**Data Science**	PRIMARY	Statistical process control, predictive maintenance, batch analytics
SE	**Software Engineering**	PRIMARY	MES and ERP integration, batch-record software, digital twin platforms
CS	**Computer Science**	SIGNIFICANT	Manufacturing execution systems, machine-vision inspection, digital thread
CE	**Computer Engineering**	SIGNIFICANT	Robotic cell control, PLCs, embedded quality instruments
EE	**Electrical Engineering**	SIGNIFICANT	Factory power, motor drives, automation and sensing networks
MSE	**Materials Science & Eng.**	SIGNIFICANT	Process-structure-property, qualification, single-use bioprocess materials
CTP	**Cybersecurity Tech. & Policy**	SIGNIFICANT	OT and industrial control system security, GMP data integrity

Source: ECS 1100 degree-program by sector matrix, Sector 5 column (D. Burleson, Fall 2026)

Why Texas is building factories



Texas Instruments, Sherman

New chip fab began production Dec 2025; up to $40B planned

Samsung, Taylor

$17B chip fab, now set to build Tesla AI6 chips

SpaceX, Bastrop

Starlink production, plus a planned $16.8B chip plant

Apple, Houston

New factory building AI servers

In this job log: 71 of 100 postings are in Texas, and 21 are in DFW. Texas Instruments, SpaceX, Samsung, Lennox, Celestica and Alcon all appear.

Source: TI newsroom (Dec 17, 2025); Samsung (2021) and Tesla (2025); Office of the Texas Governor (Oct 1, 2026); Apple newsroom. Photo: "Semiconductor clean room" by Aileen Devlin/Jefferson Lab (public domain), Wikimedia Commons

Texas biomanufacturing is growing too



Eli Lilly, Houston

$6.5B medicine plant

Bristol Myers Squibb, Houston

About $2.3B investment

FUJIFILM, College Station

$300M expansion

Lonza, Pearland

300,000 sq ft cell and gene therapy site

Alcon, Fort Worth

$186M expansion (proposed)

Texas A&M NCTM

Training center, with a Dallas site at Pegasus Park

BMEN takeaway: these sites hire process, validation and quality engineers, the same titles you will find in the BMEN section of the job log.

Source: Office of the Texas Governor (Oct 1, 2026); FUJIFILM Diosynth; Lonza; Texas A&M NCTM; Alcon (accessed Nov 1, 2026). Photo: lab technician at a biosafety hood, U.S. Navy photo by Tom Watanabe (public domain)

What is exciting right now



1 Humanoid robots

Austin startup Apptronik raised more than $935M to build humanoid robots for factories.

2 A Texas chip fab goes live

TI's new Sherman fab began production in Dec 2025.

3 Robots building robots

Tesla says its Giga Texas Optimus line is designed for up to 10M robots a year (a target).

4 Digital twins

Software copies of a factory, used to test changes before making them.

Source: Apptronik (Feb 11, 2026); TI (Dec 17, 2025); Teslarati on Giga Texas Optimus plans (company target); NIST (Feb 13, 2026). Photo: humanoid robot at Science Square Tsukuba, by Syced (CC0), Wikimedia Commons

Manufacturing at UT Dallas



Manufacturing lab (CAML)

Machining, 3D printing and smart manufacturing research (Dr. Wei Li, ME).

Cleanroom research

NSERL cleanroom; $700K state grant (Apr 2026) for new tools.

Biomedical Device Center

Takes medical devices from idea to prototype.

Supply chain at Jindal

Ranked #8 by Gartner (2022). Good for an elective or minor.

Student organizations

Comet Robotics | RoboSub | ASME | Dallas Formula Racing | SAMPE | BMES | Biotech Club

Interested in research? Email a professor whose lab works on something in this deck.

Source: UT Dallas news and department pages (accessed Nov 1, 2026); Texas Semiconductor Innovation Fund (Apr 10, 2026). Photo: NSERL building, UT Dallas, by Stan9999 (public domain), Wikimedia Commons

When one stage fails, the whole system feels it

Chip fabs at capacity → **Auto makers short on chips** → **Lines stop, half-built trucks parked** → **Dealer lots empty, prices rise**

27.3% of transportation-equipment makers cited not enough materials as the reason they could not produce more

3.6M → 1.5M U.S. vehicle inventory fell as the shortage spread

Same logic in biomanufacturing: one contaminated bioreactor or one failed release test can stop a batch of medicine. When you build your concept map, ask of every node: what breaks downstream if this one fails?

Source: Federal Reserve Bank of Cleveland, Economic Commentary on the 2021 semiconductor shortage (accessed Nov 1, 2026)

Two decisions on the factory floor

Case A: automation

Your automation project will remove a full shift of operators. It also takes people away from a hazardous task. What do you owe the workers whose jobs change?

Case B: the batch record

The release deadline is tonight. Testing is incomplete, and you are asked to sign the GMP batch record anyway. Who would find out? What does your signature say?

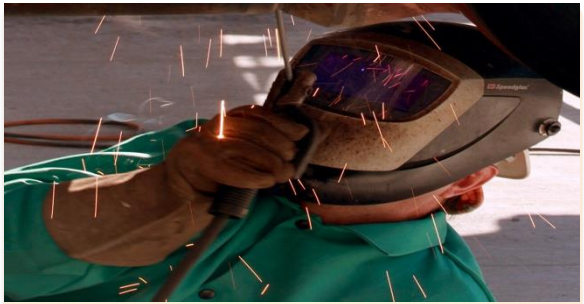
Why it matters

- FDA expects records to be ALCOA: attributable, legible, contemporaneous, original and accurate
- In Jan 2025, the FDA sent a warning letter (320-25-30) to a drug maker for falsified manufacturing records
- NSPE Canon 1: engineers hold public safety, health and welfare paramount

What does your signature commit you to in your role?

Source: ECS 1100 Sector 5 ethics case; FDA, Data Integrity and Compliance with Drug CGMP guidance (Dec 2018); FDA Warning Letter 320-25-30 (Jan 16, 2025); NSPE Code of Ethics, Canon 1

Worker safety is an engineering decision



2.7 injuries per 100 workers

332,600 cases

U.S. manufacturing, 2024

1. **Design out the hazard**
   Guards, interlocks and robots for the dull, dirty and dangerous tasks

2. **Automate with people in mind**
   Retrain and redeploy, not only remove

3. **Count safety in the numbers**
   44 of the 100 postings in our log ask for safety skills (PPE, OSHA, EHS)

4. **Speak up**
   Stop-work authority means anyone can halt an unsafe line

Source: U.S. BLS, Survey of Occupational Injuries and Illnesses, 2024 data (accessed Nov 1, 2026). Photo: welder in protective gear, NAVFAC (CC BY 2.0), Wikimedia Commons

How the job log was built

01

Search

LinkedIn, posted in the past 30 days; Texas first, then U.S.

02

Screen

The work must be in manufacturing or supply chain, and the posting must accept the major

03

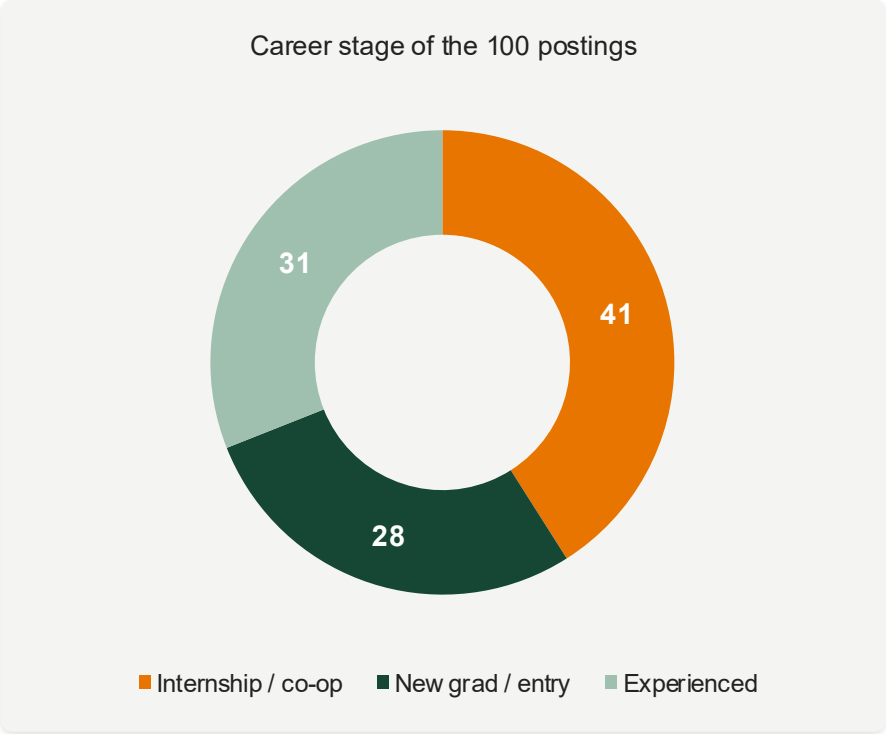
Verify

Each posting opened and confirmed open (Nov 1 or Nov 8, 2026); 10 per major

04

Archive

Full text saved as a PDF, so the job can be shown after it closes



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

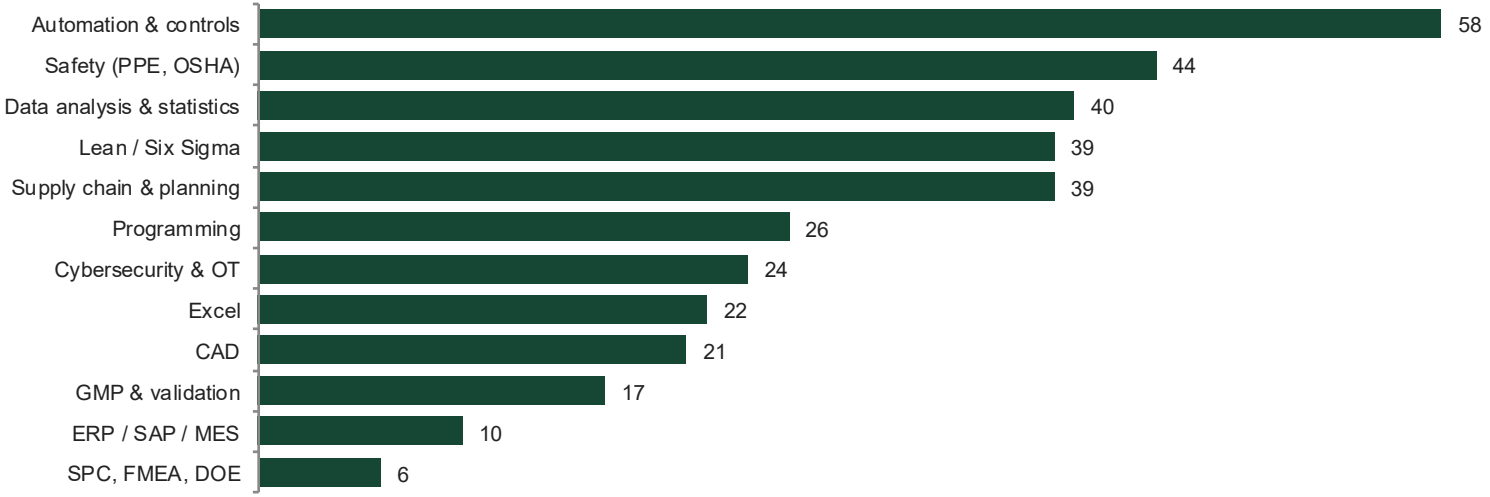
What almost every posting asks for

80%
Teamwork and collaboration

80%
Communication, written and verbal

67%
Problem solving and root cause

58%
Automation and controls



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026; a skill counts once per posting

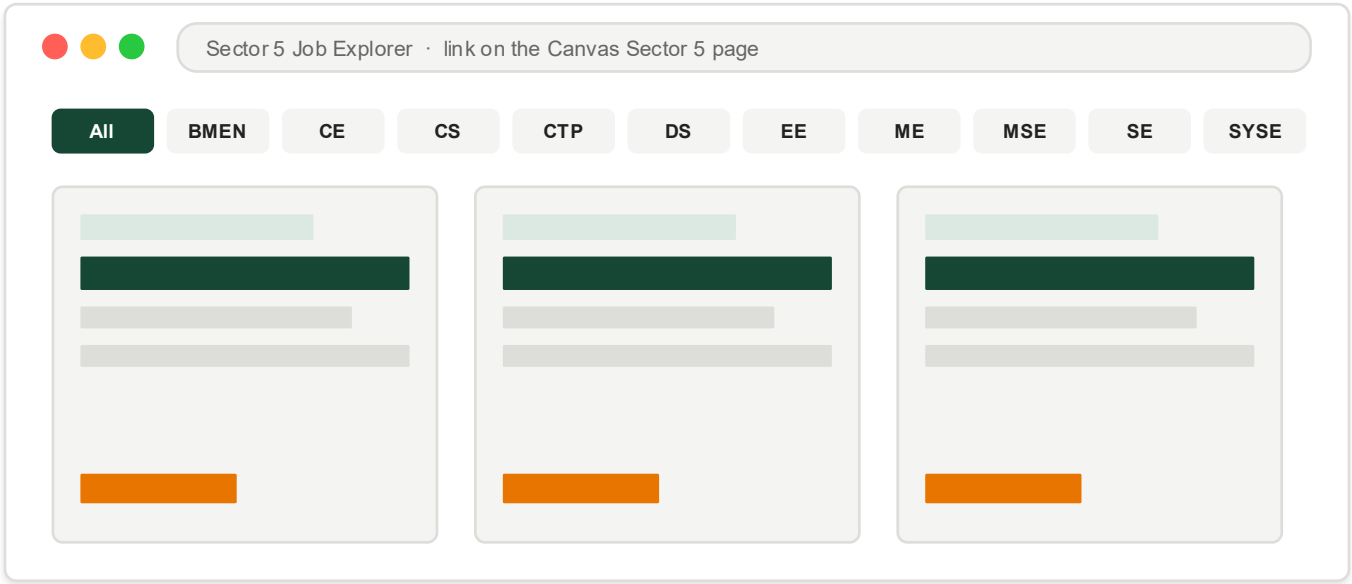
Top technical asks by major

BMEN	GMP & validation 8 · Safety 4 · Data analysis & statistics 4
CE	Automation & controls 10 · Lean / Six Sigma 6 · Safety 4
CS	Automation & controls 8 · Programming 7 · Data analysis & statistics 5
CTP	Automation & controls 8 · Cybersecurity & OT 8 · Safety 4
DS	Supply chain & planning 9 · Data analysis & statistics 9 · Lean / Six Sigma 6
EE	Safety 8 · Automation & controls 6 · CAD 4
MSE	Supply chain & planning 7 · Safety 5 · Data analysis & statistics 3
ME	Safety 6 · Lean / Six Sigma 6 · Automation & controls 6
SE	Automation & controls 9 · Programming 7 · Data analysis & statistics 4
SYSE	Data analysis & statistics 6 · Lean / Six Sigma 6 · Excel 6

Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026; counts out of 10 postings per major; teamwork and communication left out

Where to find them all

Every posting behind this deck, in one place



Filter by major, career stage, sub-sector or Texas. Each card links to the live LinkedIn posting and to an archived PDF of the full text.

Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Career support on campus

Jonsson School Career Services

One-on-one help with resumes, LinkedIn profiles and internship searches.

University Career Center

Job board, career fairs and employer info sessions for all majors.

UTDesign capstone

Senior design projects sponsored by companies, solving their real problems.

LinkedIn Learning (free)

Courses on PLCs, SolidWorks, Excel, Python and lean, free through the library with your NetID.

Try this week: open three postings for your major in the explorer, list the skills you do not have yet, and match each one to a class, a lab, an org or a free course.

Source: UT Dallas Jonsson School Career Services, University Career Center, UTDesign and McDermott Library pages (accessed Nov 1, 2026)

Questions to bring to the panel

1. Which stage of the production system do you work in, and which majors do you work with every day?
2. What happened the last time one stage failed, and how did your team find out?
3. How has automation changed the jobs on your floor in the past five years?
4. What does a first-year engineer do in your company in month one?
5. Have you ever had to refuse to sign something? What happened?

Source: ECS 1100 Sector 5 outcomes and ethics case

HW 12: Sector 5 concept map

1. **Map both branches**

   Discrete manufacturing and biomanufacturing, from design intent to delivery

2. **Add the discipline layer**

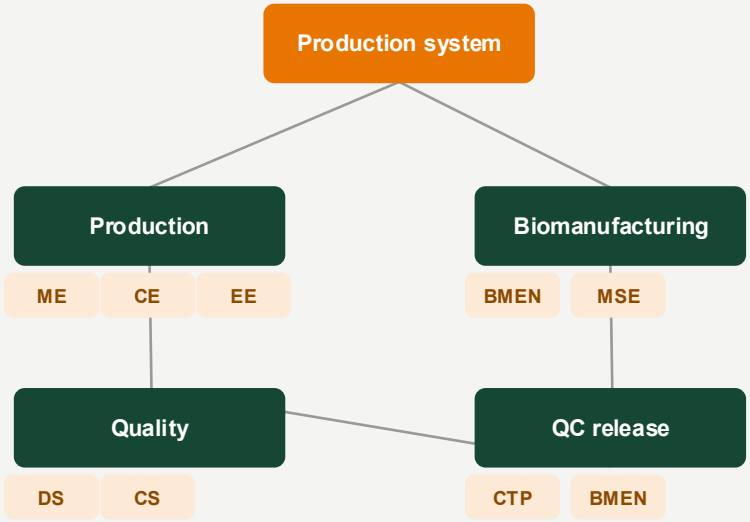
   Label every node with the degree programs acting on it

3. **Find the busiest node**

   Which node carries the most programs? Which carries the fewest, and is that real or a gap in your map?

4. **Link the ethics case**

   Connect the batch-record decision to the quality node it affects



Example only: your map will have many more nodes.

HW 12: submit your concept map on Canvas by Sun, Nov 15, 11:59 PM. Use the video, these slides and the panel.

Source: ECS 1100 Guidance for Sector Concept Maps (D. Burleson, Fall 2026)

APPENDIX

Major guides

One page per Jonsson School program, drawn from the 100-posting job log

Biomedical Engineering (BMEN)

10	5	1	4
postings	internships / co-ops	new grad / entry	in Texas

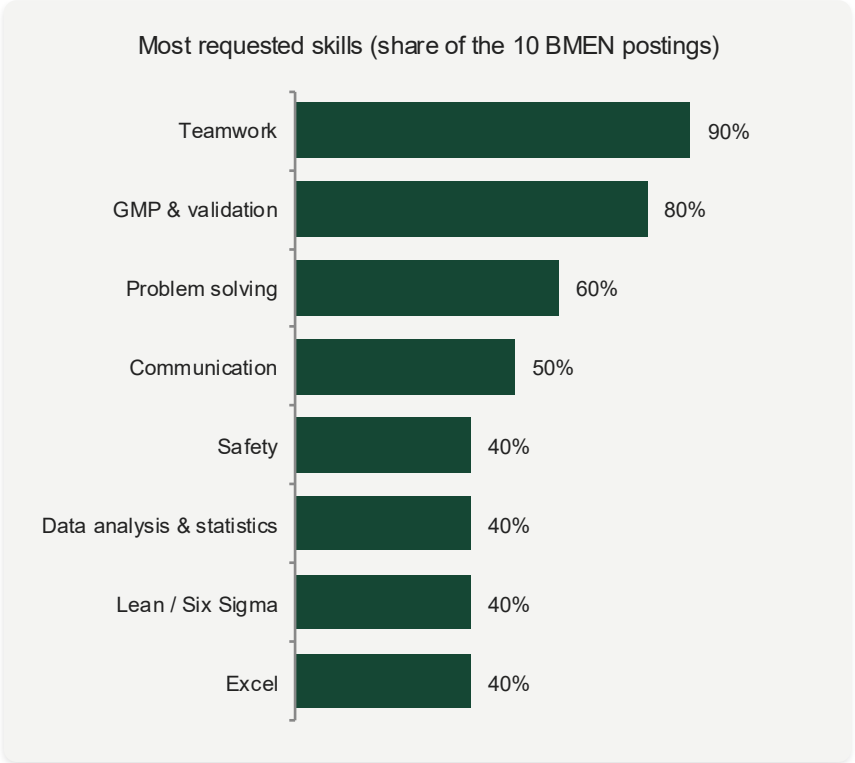
PRIMARY role in this sector

Biomanufacturing: upstream culture, downstream purification, fill-finish, sterile process and GMP release.

Job titles seen

- 2027 Intern - Process Development, Downstream (Catalent)
- Validation Engineer (W) (Cellipont Bioservices)
- Process Engineer II (Curia)
- 2027 Validation Engineer Co-Op (July - Dec) (Under… (AbbVie)

Only 1 new-grad BMEN posting was found; entry roles often say "Engineer I" or "Associate" under process, validation or quality.



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Computer Engineering (CE)

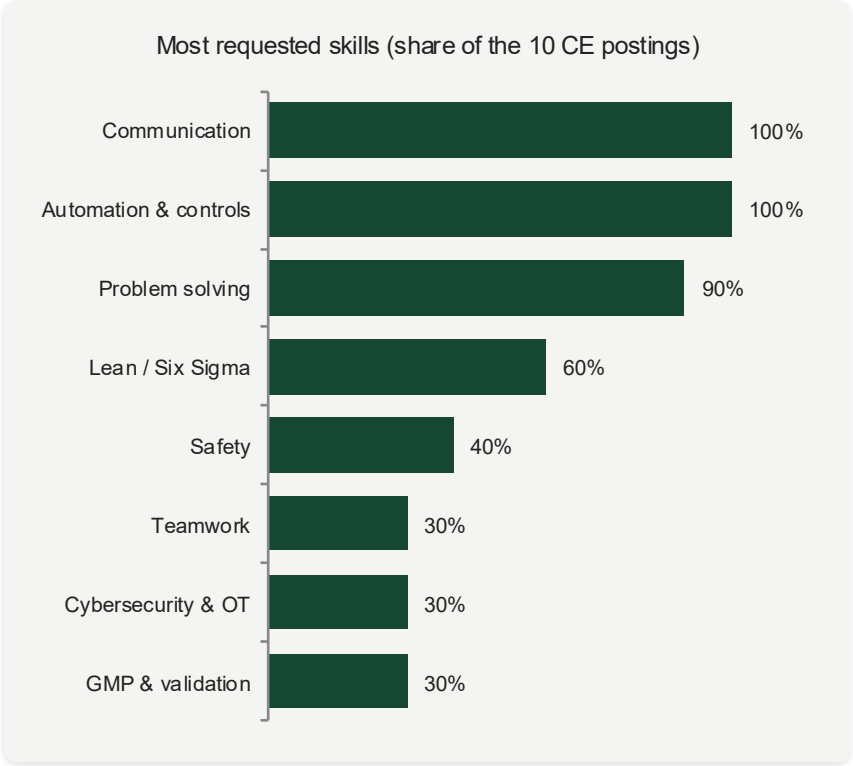
10	4	3	9
postings	internships / co-ops	new grad / entry	in Texas

SIGNIFICANT role in this sector

Robotic cell control, PLCs and embedded quality instruments on the line.

Job titles seen

- Instrumentation/Automation Engineering Intern (Toshiba International Corporation)
- Test Engineer (Delta Electronics Americas)
- Sr. Automation Engineer (Alcon)
- Systems Integration Intern (Emerson)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Computer Science (CS)

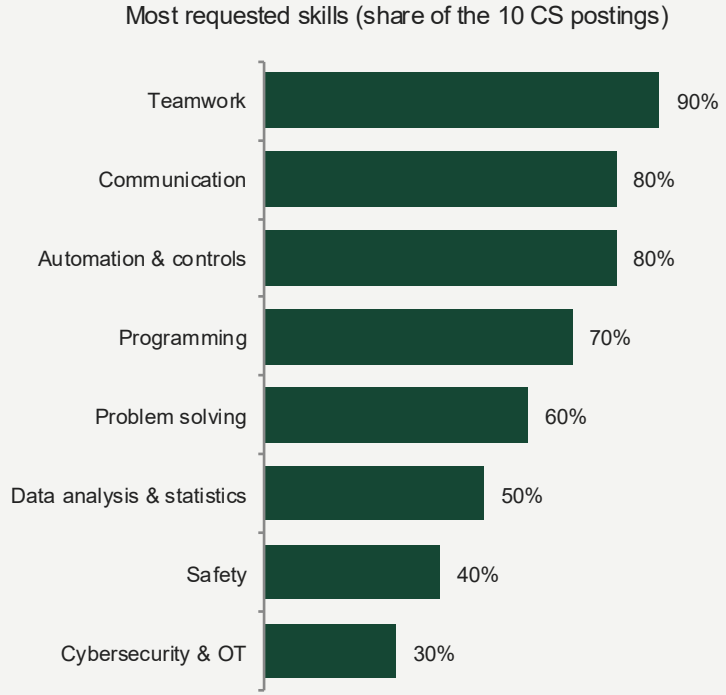
10	4	3	5
postings	internships / co-ops	new grad / entry	in Texas

SIGNIFICANT role in this sector

Manufacturing execution systems, machine-vision inspection and the digital thread that tracks each part.

Job titles seen

- Modernization Intern (Emerson)
- Factory Software Engineer (Starlink) (SpaceX)
- Fab Automation and Software Engineer (TSMC)
- Automation Engineering Intern (Novelis)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Cybersecurity Technology & Policy (CTP)

10	4	3	5
postings	internships / co-ops	new grad / entry	in Texas

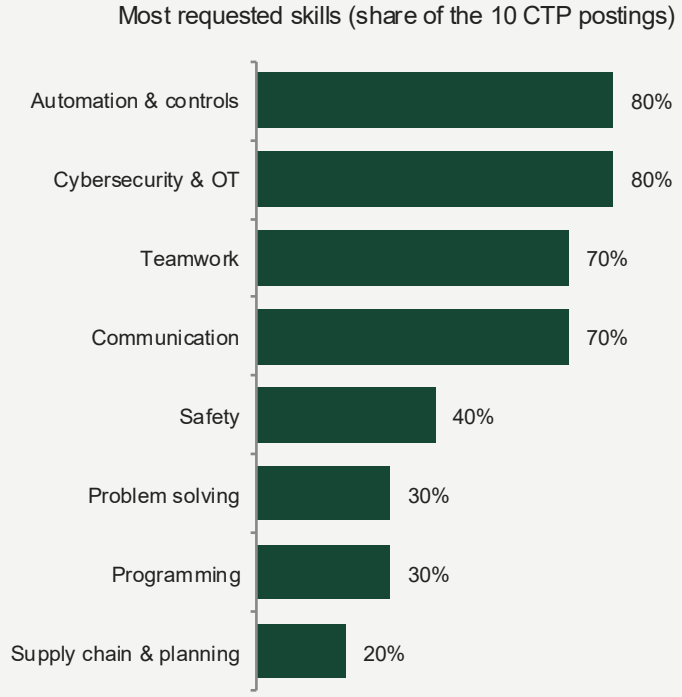
SIGNIFICANT role in this sector

Securing factory networks (OT) and industrial control systems; GMP data integrity and compliance.

Job titles seen

- Cyber Resilience & Product Security Intern (DwyerOmega)
- Product Security Engineer (Starlink) (SpaceX)
- OT Cybersecurity Engineer (Vulcan Elements)
- Early Career Internship Program: Information Techn… (Eaton)

Many OT security roles ask for some experience; an IT or controls internship is the usual first step.



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Data Science (DS)

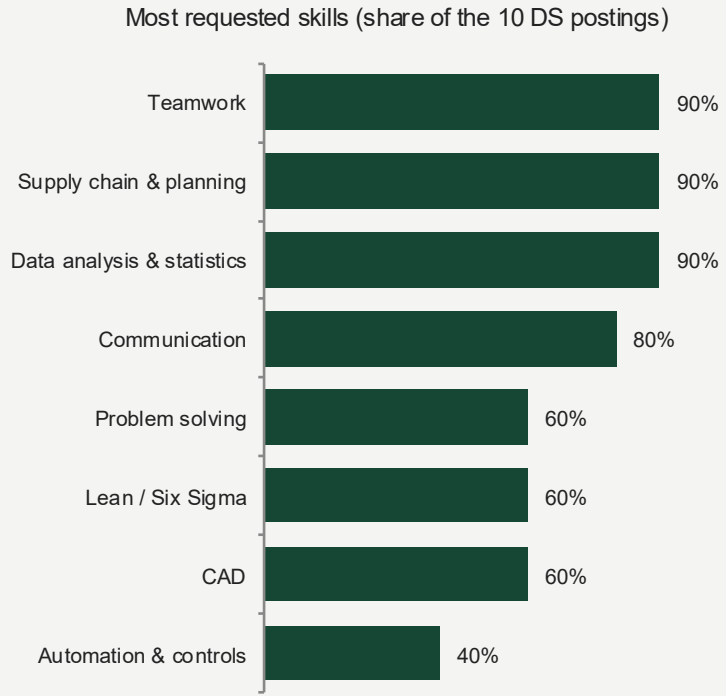
10	4	3	9
postings	internships / co-ops	new grad / entry	in Texas

PRIMARY role in this sector

Statistical process control, predictive maintenance, demand planning and batch analytics.

Job titles seen

- Demand Planning Intern (7-Eleven)
- Supply Chain Analyst I - Fresh (H-E-B)
- Advanced Analytics Research Scientist - Industrial… (Rockwell Automation)
- Winter 2027 Co-op - Quality Data Analytics (Keurig Dr Pepper Inc.)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Electrical Engineering (EE)

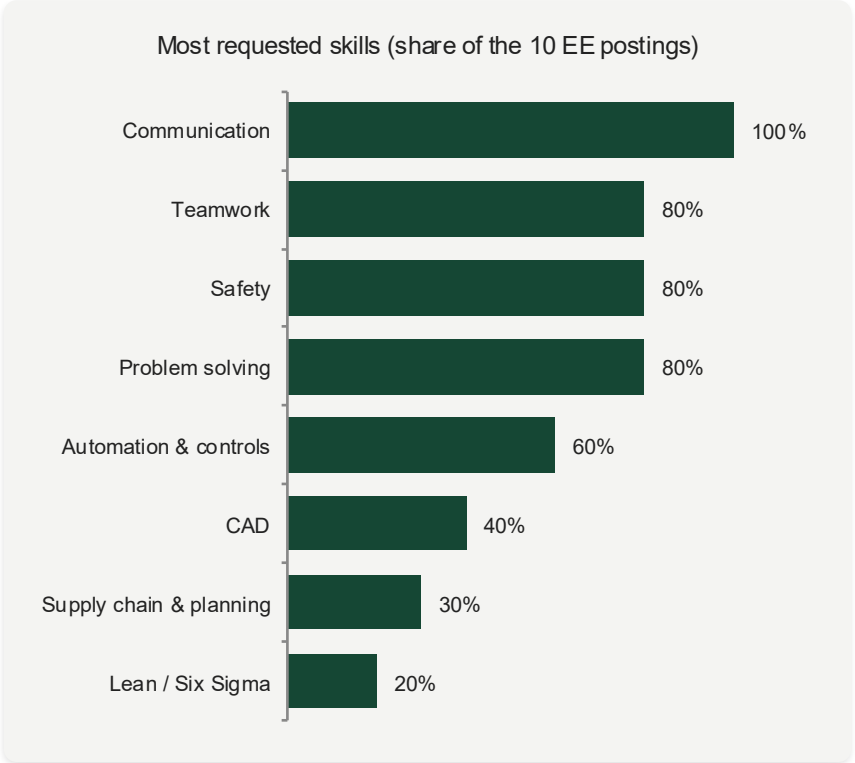
10	4	3	10
postings	internships / co-ops	new grad / entry	in Texas

SIGNIFICANT role in this sector

Factory power, motor drives, controls, and automation and sensing networks.

Job titles seen

- I&E Engineer Intern (Indorama Ventures: Indovinya)
- Controls Eng Technician II (General Dynamics Ordnance and Tactical Systems)
- Systems Controls Engineer (Grainger)
- Engineering Intern I (Graphic Packaging International)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Materials Science & Engineering (MSE)

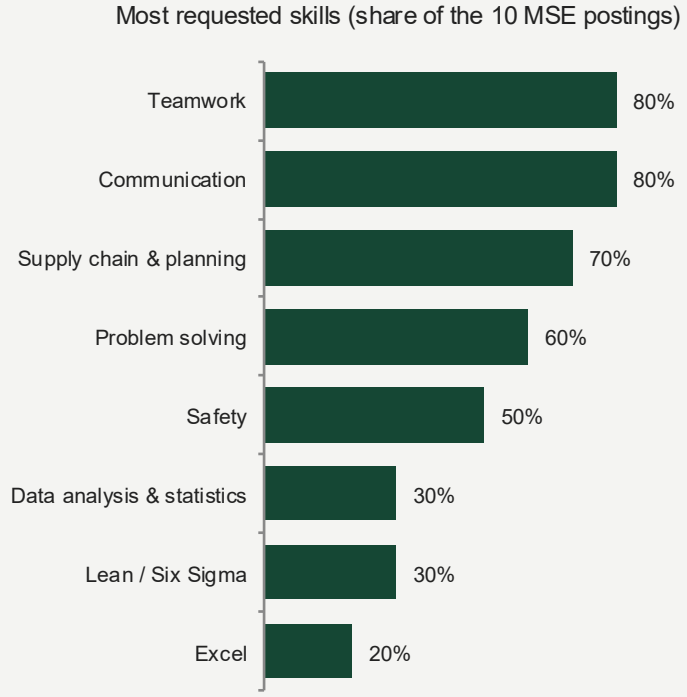
10	4	3	7
postings	internships / co-ops	new grad / entry	in Texas

SIGNIFICANT role in this sector

Process-structure-property work, materials qualification, metallurgy and single-use bioprocess materials.

Job titles seen

- 2027 Chemistry, Materials & Life Sciences Interns (Alcon)
- Materials Engineer (Starlink) (SpaceX)
- Materials Engineer (Pantex Plant)
- Mercury Marine - Heat Treat Metallurgical Engineer… (Brunswick Corporation)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Mechanical Engineering (ME)

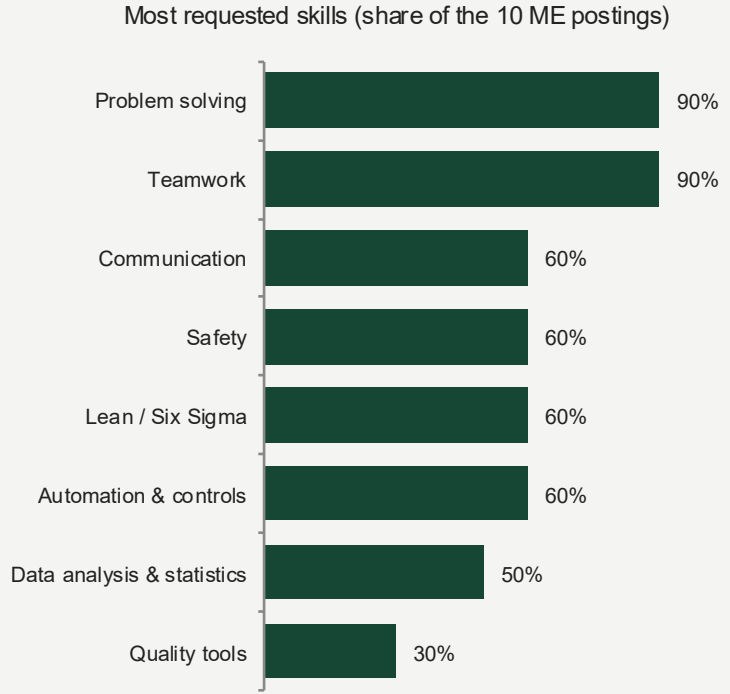
10	4	3	10
postings	internships / co-ops	new grad / entry	in Texas

PRIMARY role in this sector

Process and tooling design, additive manufacturing, and production equipment.

Job titles seen

- 2027 Manufacturing Engineer Summer Internship (Bac… (Applied Materials)
- Manufacturing Integration Engineer (Starship) (SpaceX)
- Manufacturing Engineer (Schreiber Foods)
- Summer Engineering Intern (StandardAero)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Software Engineering (SE)

10	4	3	3
postings	internships / co-ops	new grad / entry	in Texas

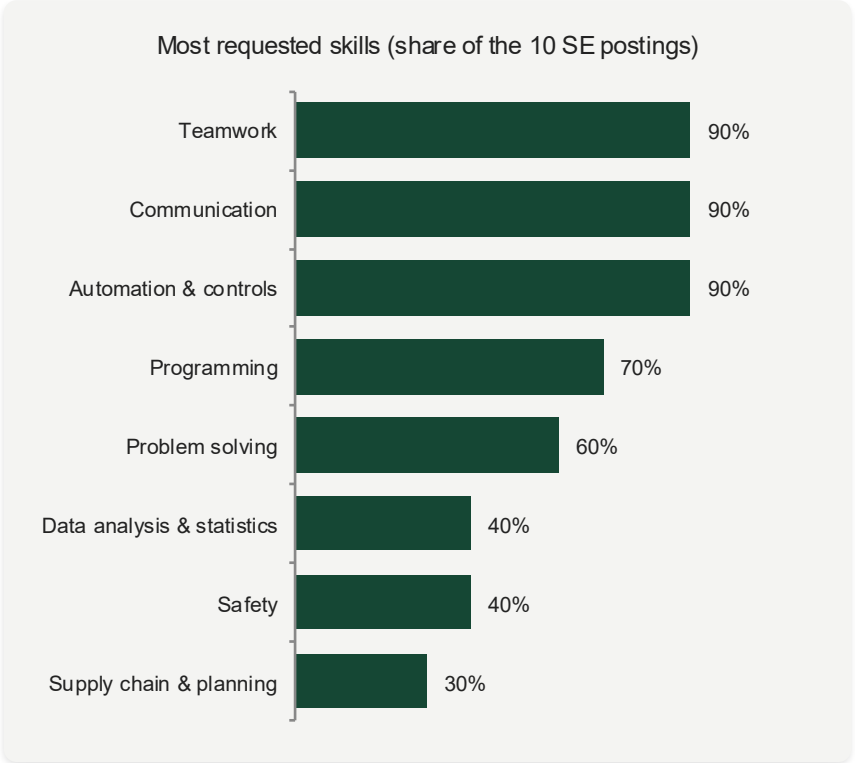
PRIMARY role in this sector

MES and ERP integration, factory software platforms, test automation and digital twins.

Job titles seen

- Intern - Product Yield Software Engineer (Micron Technology)
- Software Engineer, Robot Manufacturing (Neuralink)
- Senior Software Engineer - Forge Factory Automation (Anduril Industries)
- Software Engineer Intern - (Jan 2027) (Westmag)

Only 3 of the 10 SE postings are in Texas; factory-software roles also cluster in CA, WA and WI.



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026

Systems Engineering (SYSE)

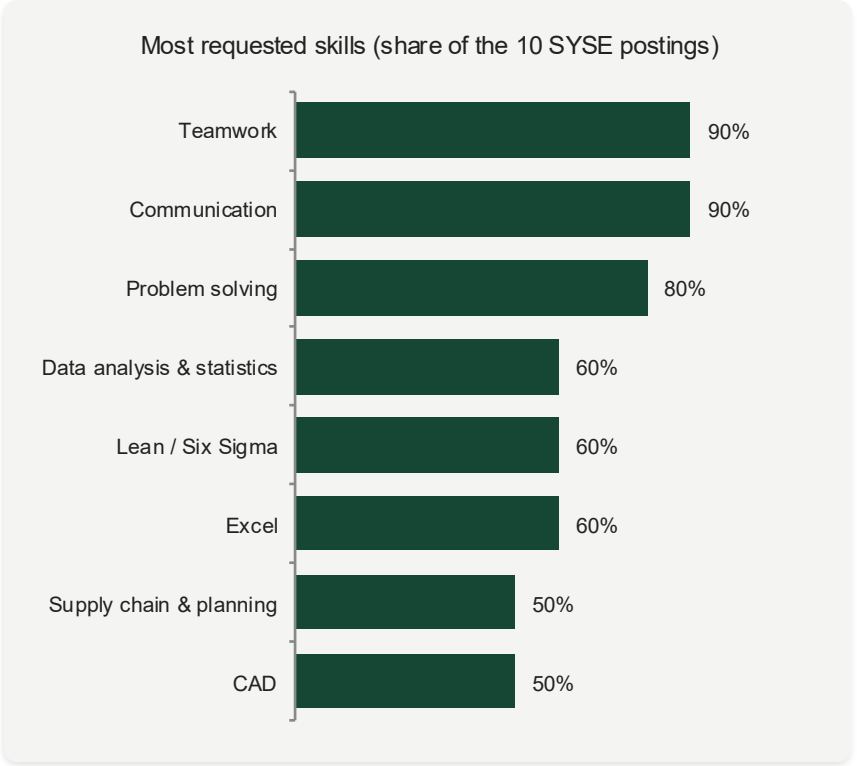
10	4	3	9
postings	internships / co-ops	new grad / entry	in Texas

PRIMARY role in this sector

Production systems, industrial engineering, logistics, quality systems and operations research.

Job titles seen

- Industrial Engineering Intern II, Summer 2027 - Au… (BAE Systems, Inc.)
- Industrial Systems Engineer - RTD (Texas Instruments)
- RFAB Module Industrial Engineer (Texas Instruments)
- Industrial Engineer - Intern (Zekelman Industries)



Source: ECS 1100 Sector 5 Job Log, 100 LinkedIn postings from 74 employers, checked Nov 1 and Nov 8, 2026